

ABSTRACT

1. Purpose:

The purpose of this project is to analyze player performance data from the T20 World Cup 2022, sourced from the ESPN website, with the goal of selecting an optimal 15-member cricket squad comprising openers, batters, bowlers, and all-rounders. Cricket team selection is often influenced by intuition or limited metrics.

2. Problem Addressed:

The problem this project aims to solve is the lack of objective, quantitative methods in forming a balanced team based on actual tournament performance. To ensure fair comparison, different metrics are used for different roles.

3. Approach:

Our approach involves extracting and cleaning data from multiple JSON files covering match summaries, player statistics, and individual performances. Players are categorized into roles and ranked based on weighted performance metrics.

4. Key Findings:

Key findings include the identification of top-performing players who were not always part of the official playing elevens, showcasing the value of unbiased, performance-based analysis.

5. Significance:

This analysis is significant as it offers a scalable framework for cricket analysts, selectors, or enthusiasts to make informed decisions. In the future, the model can be expanded to incorporate live data feeds, machine learning predictions, and adaptability.

6. Future Scope:

The project can evolve by integrating live data, applying machine learning for predictions and role classification, optimizing team combinations, and adapting the model to other cricket formats and leagues.

CHAPTER 1 INTRODUCTION

1.1 MOTIVATION

Cricket is more than just a sport—it is a passion shared by millions across the globe, especially in countries like India, Australia, England, and Pakistan. Among all formats, T20 cricket has gained immense popularity due to its fast-paced nature, unpredictable outcomes, and intense competition. With the rise of T20 leagues and tournaments such as the ICC T20 World Cup, cricket has become increasingly data-driven. Teams now rely not only on experience and intuition but also on statistical insights to evaluate player performance, devise strategies, and make informed decisions.

In the era of digitization, a vast amount of cricket data is readily available on platforms like ESPN, Cricbuzz, and ICC's official websites. However, this data often remains underutilized in academic and analytical projects. This motivated us to use real-world data from the T20 World Cup 2022 to build a data-driven solution that analyzes player performance and helps in selecting the best possible squad based on well-defined criteria.

Another motivation behind this project was to bridge the gap between raw data and decision-making. Traditional methods of player selection often rely heavily on reputation, limited sample sizes, or non-quantifiable traits. While expert opinion is valuable, it can be complemented with objective data analysis to ensure fairness and consistency. By developing a system that categorizes players based on roles such as openers, batters, bowlers, and all-rounders, and then evaluates them using performance metrics like strike rate, economy rate, wickets, and runs, we aim to promote merit-based selection.

This project also aligns well with the growing interest in sports analytics, a field that combines computer science, data analysis, and domain expertise. For students pursuing data science or computer engineering, such projects offer practical exposure to real-

world datasets, data cleaning, feature engineering, and decision-making through code. It provides an opportunity to apply concepts like filtering, sorting, aggregation, and visualization in a meaningful way.

Moreover, this project is scalable and adaptable. The same approach can be applied to different formats (ODI, Test) or tournaments (IPL, BBL), and even to other sports like football or basketball. The possibility of integrating machine learning for predictive analysis or building live dashboards opens further learning and development avenues.

In summary, the motivation for this project stems from a desire to harness data for smarter decision-making in sports. It combines our academic interest in data science with our passion for cricket, demonstrating how analytical thinking can enhance sports management and fan engagement alike.

1.2 SCOPE

This project focuses on analyzing player performance data from the ICC T20 World Cup 2022 to form an ideal 15-member squad comprising openers, middle-order batters, bowlers, and all-rounders. The scope of the project extends across data collection, preprocessing, categorization, performance evaluation, and player selection using structured, objective criteria. The central idea is to demonstrate how cricket team selection can be improved using data science techniques, ensuring fairness, transparency, and evidence-based decisions.

The primary data source used in this project is ESPN's publicly available match data, which includes detailed batting, bowling, and player information for every match in the tournament. This data is extracted in JSON format and preprocessed into structured pandas DataFrames using Python. The cleaning process involves removing noise, handling encoding issues, and mapping match and player IDs. The integrity and completeness of this data are crucial, as the entire analysis and player selection depend on accurate information.

The project covers the following key components:

Player Categorization: Players are grouped into categories—openers, middle-order batters, bowlers, and all-rounders—based on their role and performance characteristics during the tournament.

Performance Metrics: Different roles require different performance indicators. For example, batters are assessed based on strike rate and total runs, bowlers based on economy rate and wickets taken, and all-rounders based on a combined score of both batting and bowling contributions.

Top Player Selection: After computing role-specific scores, the top performers in each category are selected to form a well-balanced 15-member team. The composition aims to maintain diversity in roles while ensuring each player adds value to the squad.

Analysis and Insights: The project also involves statistical analysis and ranking to identify top contributors, potential underdog performers, and players who made significant impacts in crucial matches.

This project does not focus on match-by-match predictions or real-time updates; it is limited to post-tournament performance evaluation. The scope is also confined to players who participated in the T20 World Cup 2022 and whose data was successfully captured in the ESPN datasets.

The scope may be expanded in the future to include real-time match analytics, role classification using machine learning, player form prediction, or simulation of match outcomes using selected teams. Additionally, the framework can be reused for analyzing other international or domestic cricket tournaments with similar data structures.

Overall, this project demonstrates how data science can be applied to the sports domain, particularly cricket, to draw actionable insights and support strategic decision-making. It highlights the potential of combining technical skills with domain knowledge to create impactful, real-world applications.

1.3 OBJECTIVE

The main objective of this project is to build a data-driven approach for selecting an ideal 15-member cricket team based on the performances of players in the ICC T20 World Cup 2022. The team should comprise a balanced mix of openers, middle-order batters, bowlers, and all-rounders, chosen strictly based on their statistical performance during the tournament. Unlike traditional team selections that may be influenced by subjective opinions or past reputations, this project aims to adopt an objective and analytical method that ensures fairness and transparency in the selection process.

The first step in fulfilling this objective is to collect comprehensive and structured data. This data is sourced from ESPN's publicly available resources, which include match summaries, individual batting and bowling statistics, and player information. Since the raw data is in JSON format, it is parsed and converted into structured dataframes using Python. Preprocessing includes cleaning inconsistencies, handling encoding errors, mapping team names to match IDs, and ensuring that all performance metrics are accurate and relevant.

Once the data is cleaned and ready, players are categorized into distinct roles based on their on-field contributions. These roles include openers, batters, bowlers, and all-rounders. Each category requires a different evaluation approach. Openers, for instance, are judged primarily on their strike rate and ability to score quick runs during the powerplay overs. Middle-order batters are assessed based on consistency, average, and strike rate. Bowlers are evaluated using metrics such as economy rate, total wickets taken, and bowling average. All-rounders, who contribute in both batting and bowling, are scored by combining their performance in both departments using a weighted formula.

After role classification and metric definition, players within each category are ranked based on their performance. The highest-ranked players are shortlisted to form the final squad. The selection ensures a balanced team structure with adequate representation from each role — for example, including at least two openers, four middle-order batters, two all-rounders, and a mix of fast and spin

bowlers. This method of selection ensures that the final team is not only high-performing on paper but also well-structured for real match scenarios.

An additional objective of this project is to generate visual insights from the analysis. Using data visualization tools like Matplotlib and Seaborn, key trends, comparisons, and distributions are presented in a graphical format. These visuals help support the selection logic and highlight standout players who may not be well-known but have performed exceptionally during the tournament.

Another goal of the project is to build a flexible and scalable framework that can be reused for future tournaments like the Indian Premier League (IPL), One-Day Internationals (ODIs), or even Test matches. The modular design of the data processing and evaluation logic allows the project to be easily adapted to other data sets with similar structures.

Beyond just selecting a team, the broader aim of the project is to demonstrate the power of data analytics in sports. It encourages the adoption of technology and data science in cricket analysis, aiding coaches, selectors, analysts, and even fans in making informed decisions. This project also serves as a practical learning experience, giving students hands-on exposure to data science concepts such as data wrangling, feature engineering, role classification, and statistical evaluation — all within the engaging context of cricket.

In conclusion, the objective of this project is to bridge the gap between raw cricket statistics and intelligent team selection. By using data to evaluate player performance, the project shows how data science can contribute meaningfully to strategic decisions in the sporting world strategies. The goal is to create a system that not only meets current demands but also adapts seamlessly to future data growth and technological advancement

1.4 APPLICATION

The use of data analytics in sports, particularly in cricket, has revolutionized how teams strategize and select players. This project's approach to analyzing T20 World Cup 2022 data for team selection has several practical applications, both within the cricketing ecosystem and in the broader domain of sports analytics.

One of the most immediate applications is team selection and squad formation for coaches and selectors. By leveraging objective performance data, selectors can identify consistent performers and undervalued players who may otherwise be overlooked. This helps in building balanced squads that maximize strengths across different playing roles such as openers, middle-order batters, all-rounders, and bowlers. Data-driven selection reduces reliance on subjective biases and ensures meritocracy, which can lead to improved team performance and fairness in player opportunities.

Beyond initial squad formation, this analytical framework can be adapted to match preparation and strategy development. Understanding player strengths and weaknesses through detailed statistics enables coaches to plan batting orders, bowling rotations, and field placements more effectively. For example, identifying bowlers with lower economy rates against specific oppositions or openers who score quickly in powerplay overs can help tailor match strategies that exploit opposition vulnerabilities and optimize team chances of winning.

Another important application is in player performance monitoring and talent scouting. Franchise teams, domestic clubs, and national boards can use similar data analytics techniques to monitor player progress over time. Young or emerging players' performances can be tracked, enabling scouts to discover new talent based on consistent metrics rather than reputation alone. This helps in building future-ready teams by integrating promising players early and providing them with targeted development plans.

Fan engagement and media analysis is another area that benefits from such projects. Fans increasingly demand in-depth statistics, player comparisons, and predictive insights. Sports broadcasters and analysts can use data-driven reports and visualizations to enrich match commentary, pre- and post-match analysis, and feature stories. This

enhances viewer experience and keeps fans more connected and informed about player performances and team dynamics.

The project also holds value in fantasy sports and gaming platforms. Fantasy cricket leagues rely heavily on player statistics to help participants build winning teams. The analytical models developed here can be integrated into fantasy sports apps to recommend player selections, predict performances, and provide data-backed tips. This adds credibility and excitement for fantasy sports users, promoting engagement and retention.

In the realm of sports science and player fitness management, integrating performance data with physiological metrics can help coaches and trainers manage player workloads and prevent injuries. By correlating player performance with fitness levels, decisions regarding rest, recovery, and training intensity can be more informed and personalized.

Furthermore, the techniques and frameworks developed in this project are not limited to cricket alone. The underlying principles of data collection, cleaning, classification, and evaluation can be adapted to other sports such as football, basketball, and baseball. For example, in football, similar metrics can help analyze player roles like strikers, midfielders, and defenders, aiding in team formation and tactical decisions.

From an educational perspective, this project serves as an excellent learning tool for students and professionals interested in data science applications in sports. It provides hands-on experience with real-world sports data, covering essential skills like data preprocessing, feature engineering, statistical analysis, and visualization. It also bridges the gap between theoretical knowledge and practical application, fostering skills that are highly valued in the growing sports analytics industry.

Lastly, as cricket continues to embrace technology and data-driven insights, the application of such projects contributes to the digital transformation of sports management. Organizations can evolve from traditional decision-making processes to modern, evidence-based approaches. This not only improves the quality of team selections but also enhances overall competitiveness and professionalism in cricket.

CHAPTER 2 LITERATURE SURVEY

2.1 LITERATURE SURVEY

The application of data analytics in sports has gained significant momentum over the last decade, transforming traditional approaches to player evaluation, team selection, and game strategy. Cricket, being one of the most popular sports worldwide, has particularly benefited from this data-driven revolution. This chapter reviews existing research and methodologies related to cricket analytics, focusing on player performance evaluation, team selection algorithms, and role-specific metrics, which are relevant to the present project on analyzing the ICC T20 World Cup 2022 data.

One of the foundational works in cricket analytics involves the statistical modeling of player performance. Researchers have developed models using historical match data to evaluate players' batting and bowling performances quantitatively. For example, the Player Impact Index (PII) is a metric proposed to combine batting, bowling, and fielding contributions into a single comprehensive score for ranking players (Bradley et al., 2017). Such composite indices aim to capture a player's overall value to the team beyond just runs scored or wickets taken. This concept supports the objective of forming a balanced team by objectively comparing players with different roles and skills.

Another important area of study is the role-based evaluation of players, where players are classified into distinct roles such as openers, middle-order batters, all-rounders, and bowlers. Research by Kumar and Sharma (2019) demonstrated that role-specific metrics provide better insight than aggregate statistics. For example, openers are analyzed for their ability to score quickly in powerplays, while middle-order batters are judged on consistency and ability to stabilize innings. Bowlers are evaluated on economy rates and strike rates depending on their bowling style. This approach aligns with the current project's methodology of selecting players based on role-specific performance criteria.

The use of machine learning models in player performance prediction and selection has also been explored. Techniques like decision trees, random forests, and neural networks have been applied to cricket datasets to predict player performance in upcoming matches (Singh & Jain, 2020). Although predictive modeling goes beyond the scope of this project, the data preprocessing and feature engineering techniques from these studies inform how to handle cricket data effectively. Furthermore, machine learning aids in uncovering hidden patterns and player form trends that are not obvious through simple statistics.

Several studies have addressed the challenge of balancing a cricket team based on multiple constraints such as player roles, budget, and team composition rules. For instance, the work by Patel and Desai (2018) proposed optimization models that select the best squad from a pool of players by maximizing overall performance metrics while maintaining required team structure. Such optimization approaches highlight the importance of balanced team composition — a key aspect in this project's team formation of 15 players across different roles.

The T20 format, in particular, has unique demands that differ from longer formats like ODIs or Test cricket. Studies focused on T20 cricket emphasize strike rate, boundary percentage, and economy rate as crucial metrics due to the fast-paced nature of the game (Jones et al., 2021). The impact of powerplay overs and death overs also affects player evaluation criteria. This specificity in metric selection ensures that players suited for T20 cricket are favored, which is critical for forming a team tailored for the World Cup 2022 format.

Publicly available cricket data sources such as ESPN Cricinfo's API and Kaggle datasets have been widely used in cricket analytics research. Several open-source projects and research papers have demonstrated methods for extracting, cleaning, and structuring cricket data from these sources (Patel et al., 2020). This project follows similar data processing pipelines — loading JSON data, mapping player and match identifiers, and cleaning textual data — ensuring accuracy and consistency for analysis.

Visualization of cricket data is another important research theme. Visual tools such as heat maps, run-rate graphs, and wicket distributions help in intuitive understanding of player performance and match situations (Smith & Taylor, 2019). Visualization supports decision-making by presenting complex statistics in an easily digestible format for selectors and coaches. In this project, data visualization techniques are used to highlight top performers and compare players across roles, thereby facilitating transparent team selection.

In the domain of sports analytics, the broader significance of using data for talent identification and player scouting has been demonstrated. Studies on scouting underline the importance of performance trends over multiple seasons and matches rather than isolated performances (Lee & Park, 2020). This reinforces the need for comprehensive datasets and consistent performance evaluation metrics, as applied in this project to ensure the selected players are truly among the best in their roles.

Despite many advances, challenges remain in cricket analytics. The variability of playing conditions, opposition strength, and match contexts often complicate pure statistical analysis (Garcia et al., 2018). Researchers suggest incorporating contextual factors and advanced metrics like player pressure index or clutch performance scores for deeper insights. While this project focuses on key performance statistics, future extensions could integrate such contextual analytics for more refined player evaluation.

In summary, the literature reveals that a combination of role-specific metrics, composite performance indices, data preprocessing, visualization, and optimization techniques form the backbone of modern cricket analytics. The current project builds upon these foundations by applying a structured, data-driven approach to select a balanced T20 World Cup 2022 team. The methodologies adopted align with best practices identified in academic and industry research, validating the approach's relevance and potential impact in real-world cricket analytics.

2.2 CONCLUSION

The review of existing literature on cricket analytics and team selection highlights the growing importance of data-driven approaches in modern sports management. Over the years, researchers and practitioners have increasingly focused on developing objective and quantifiable methods to evaluate player performances, moving away from purely subjective judgments. This shift has enabled more transparent, consistent, and fair decision-making processes in selecting teams and planning strategies.

From the studies examined, it is evident that player evaluation must be role-specific to accurately reflect the varied responsibilities within a cricket team. Metrics such as strike rate, batting average, economy rate, and wickets taken are more meaningful when interpreted in the context of whether a player is an opener, middle-order batter, bowler, or all-rounder. This insight directly informs the approach taken in this project, ensuring that player assessments are nuanced and suited to their specialized roles, especially within the dynamic T20 format.

The literature also confirms the utility of advanced statistical models and machine learning algorithms in predicting player performance and optimizing team composition. While predictive modeling is not the primary focus of this project, the data processing techniques and performance criteria derived from such research have been instrumental in structuring a reliable framework for player selection based on past tournament data. Moreover, optimization models discussed in prior work provide guidance on balancing player roles while maximizing team strength, a principle central to assembling a well-rounded squad.

Visualization emerges as another critical component in cricket analytics. Effective visual representation of player statistics and comparisons aids selectors, coaches, and fans in understanding performance trends and making informed decisions. This project incorporates such visualization techniques to complement numerical analysis, making the insights more accessible and actionable.

Despite the advancements, challenges remain due to factors like match conditions, opposition quality, and player form fluctuations, which are difficult to quantify fully. The literature suggests that future research should focus on incorporating contextual and situational variables to enhance predictive accuracy and selection reliability.

In conclusion, the comprehensive review validates the use of role-specific, data-driven evaluation metrics and balanced team selection methodologies, which form the foundation of this project. By applying these well-established principles to the T20 World Cup 2022 data, this project aims to contribute to the evolving field of cricket analytics and demonstrate the practical benefits of analytics in sport team management.

CHAPTER 3 PROBLEM STATEMENT

3.1 PROBLEM STATEMENT

Build the unbeatable XI — where every player is a game-changer!

1. The team should be able to score atleast 180 runs on avg.
2. they should be able to defend 150 runs on average.

Introduction

Cricket, especially in the fast-paced T20 format, demands meticulous team selection to ensure the right balance of players who can perform under pressure. The ICC T20 World Cup 2022 showcased some of the best talents globally, with players excelling in different roles such as batting, bowling, and all-round capabilities. Selecting a top-performing team from a large pool of players requires analyzing extensive match and player data to identify those with consistent and impactful performances.

The Challenge

Traditional selection methods often rely on subjective opinions, past reputations, or limited statistical evaluation, which may overlook emerging talent or role-specific contributions. The challenge lies in objectively evaluating diverse players based on multiple performance metrics tailored to their roles — for example, distinguishing openers from middle-order batters, or death-over bowlers from spinners.

Additionally, assembling a balanced team of 15 players that meets strategic needs while maximizing overall performance is complex and data-intensive.

Objective

The primary objective of this project is to develop a systematic, data-driven framework that analyzes player performance from the T20 World Cup 2022 to select a balanced team of 15 players. This includes categorizing players into roles such as openers, batters, bowlers, and all-rounders and applying specific criteria to rank and pick top performers. The goal is to minimize subjective bias and leverage comprehensive statistical insights for better team formation.

Proposed Approach

The approach involves gathering detailed match and player data from reliable sources, cleaning and preprocessing this data, and defining role-specific performance metrics. Using these metrics, players are evaluated and ranked within their categories. Finally, a team is formed by selecting the highest-ranked players in each role, ensuring the right balance and diversity of skills. Visualizations and summary statistics support transparent and explainable decision-making.

Impact

This project demonstrates the power of sports analytics in enhancing team selection accuracy and fairness. By adopting a data-centric method, cricket teams can optimize player utilization, improve match outcomes, and discover new talent. Moreover, this framework can be extended to other tournaments and sports, fostering evidence-based decision-making and advancing the field of sports analytics.

CHAPTER 4 MINIMUM HARDWARE AND SOFTWARE REQUIREMENTS

4.1 MINIMUM SOFTWARE REQUIREMENT

1. Operating System:

- Windows, macOS, or Linux (any modern version of these OS will work)

2. Programming Language:

- Python 3.6 or above (latest stable version recommended)

3. Libraries:

- Pandas (for data manipulation and analysis)
- Scikit-learn (for TF-IDF vectorization and cosine similarity calculations)
- Numpy (for numerical operations, if needed)

4. Development Environment:

- IDE/Code Editor:
- Any of the following:
- VS Code
- PyCharm
- Jupyter Notebook (for an interactive development environment)
- Sublime Text (optional)

5. Python Package Manager:

- pip or conda (to install and manage Python packages)

4.2 MINIMUM HARDWARE REQUIREMENTS

1. Processor:

- Minimum: Intel Core i3 (or equivalent AMD processor)
- Recommended: Intel Core i5 or higher (for smoother processing, especially with larger datasets)

2. RAM:

- Minimum: 4 GB of RAM
- Recommended: 8 GB of RAM (especially for handling larger datasets and smoother performance).

3. Storage:

- Minimum: 5 GB of free disk space
- Recommended: 10 GB or more (if the movie dataset is large or if additional datasets are added) .

4. Graphics:

- Minimum: Integrated graphics (e.g., Intel HD Graphics)
- Recommended: Dedicated graphics (only if you plan to use GPU-intensive tasks, which is unlikely for this project).

5. Network (optional, for web deployment):

- Basic internet connection (if you plan to download datasets or deploy the application online).

These requirements will ensure your movie recommendation system can run smoothly without any issues. For larger datasets or additional features, you may need to scale up the hardware (RAM and storage).

CHAPTER 5 METHODOLOGY USED

5.1 METHOD

The methodology adopted in this project focuses on a structured, data-driven approach to analyze player performances from the ICC T20 World Cup 2022 and select a balanced team of 15 top players. The process involves several key steps, including data collection, preprocessing, role classification, metric formulation, player ranking, and team formation. Each step is designed to ensure the accuracy, relevance, and fairness of player evaluation while maintaining clarity and reproducibility.

Here's how the methodology works, described in detail:

Data Collection

The first step involves gathering comprehensive match and player data from reliable sources, specifically ESPN's official datasets for the T20 World Cup 2022. The data comprises match summaries, batting and bowling statistics, and player information, all provided in JSON format. These datasets contain details such as runs scored, balls faced, dismissal status, wickets taken, overs bowled, and player roles, essential for performance analysis..

Data Preprocessing

Raw data requires cleaning and structuring to be useful for analysis. This stage includes converting JSON files into structured Pandas DataFrames, renaming columns for clarity, and handling missing or inconsistent data entries. Textual inconsistencies, Such as encoding errors in player names, are corrected to ensure uniformity. Matches are identified using unique match IDs, and players are mapped to their respective teams and roles. This step also involves creating new columns, such as "out/not_out" status for batsmen, to enrich the dataset.

Role Classification

Players are categorized into specific roles—openers, middle-order batters, bowlers, and all-rounders—based on their playing position and performance patterns. This classification is vital because each role demands different skills and performance criteria. For example, openers are evaluated for their ability to provide quick starts, whereas bowlers are judged based on economy rates and wicket-taking ability. This role-specific approach enables more precise and meaningful comparisons.

Metric Formulation and Player Evaluation

For each role, customized performance metrics are formulated. Batting metrics include average runs scored, strike rate, and consistency (frequency of scoring above a threshold). Bowling metrics cover economy rate, strike rate (balls per wicket), and wickets per match. All-rounders are assessed by combining batting and bowling performances using weighted scores. These metrics are computed using aggregated match data, ensuring statistical significance.

Player Ranking and Team Formation

Players within each role are ranked according to their calculated performance scores. A balanced team is then formed by selecting the top performers from each category, ensuring the right mix of openers, batters, bowlers, and all-rounders. Constraints such as the total number of players per role are maintained to mirror realistic team compositions. This selection is guided by data rather than subjective judgment, promoting transparency and fairness.

Interactive User Input

The final step includes visualizing player rankings and performance trends using charts and graphs. These visual tools aid in validating the selections and provide insights into player contributions. The methodology's effectiveness is evaluated by comparing the selected team's profiles against actual team compositions and performances from the tournament.

5.2 ALGORITHM USED

The core of this project revolves around a systematic algorithm designed to analyze cricket player data from the ICC T20 World Cup 2022 and select a balanced team of 15 players based on role-specific performance criteria. The algorithm combines data preprocessing, classification, performance metric computation, ranking, and selection in a logical sequence. Below is a detailed description of the algorithm's steps.

Step 1: Data Loading and Preprocessing

The algorithm begins by loading JSON data files containing match summaries, batting statistics, bowling statistics, and player information. These files are parsed and converted into structured data frames for easy manipulation. Preprocessing involves cleaning data by removing or correcting corrupted characters in player names, handling missing values, and mapping match descriptions to unique match IDs to ensure consistency across datasets.

Step 2: Player Role Classification

Each player is classified into one of four primary roles: opener, batter, bowler, or all-rounder. This classification is based on the player's typical batting order position and bowling activity recorded during the tournament. For example, players who frequently bat in the first two positions are tagged as openers. Players with significant batting contributions but no or minimal bowling are categorized as batters, whereas those who bowl regularly are tagged as bowlers. Players contributing substantially in both batting and bowling are classified as all-rounders.

Step 3: Define Role-Specific Performance Metrics

Role-specific metrics are defined to evaluate player effectiveness:

Openers and Batters: Metrics include batting average (total runs scored divided by number of innings dismissed), strike rate (runs scored per 100 balls), and consistency (number of innings with runs above a set threshold).

Bowlers: Metrics include bowling economy rate (runs conceded per over), strike rate (balls bowled per wicket taken), and average wickets per match.

All-rounders: A weighted score combining batting average and bowling economy is computed to reflect overall contribution.

Step 4: Compute Performance Scores

For each player, the relevant metrics are calculated from the aggregated match data. Missing or incomplete data points are handled carefully to avoid bias. For example, players who have fewer innings are assessed with adjusted averages to reduce overestimation.

Step 5: Ranking Players Within Roles

Players within each role are ranked based on their computed performance scores. Different roles have tailored ranking formulas that emphasize the most relevant metrics — for example, strike rate and consistency carry more weight for openers, while economy rate is prioritized for bowlers. The ranking ensures fair comparison among players fulfilling the same role.

Step 6: Team Selection

The algorithm selects the top players from each role category to form a balanced team of 15 members. The selection criteria typically follow standard team composition rules: for example, 3-4 openers, 5-6 batters, 4-5 bowlers, and 2-3 all-rounders, but these numbers can be adjusted based on tournament strategy. This step ensures that the team has both depth and versatility.

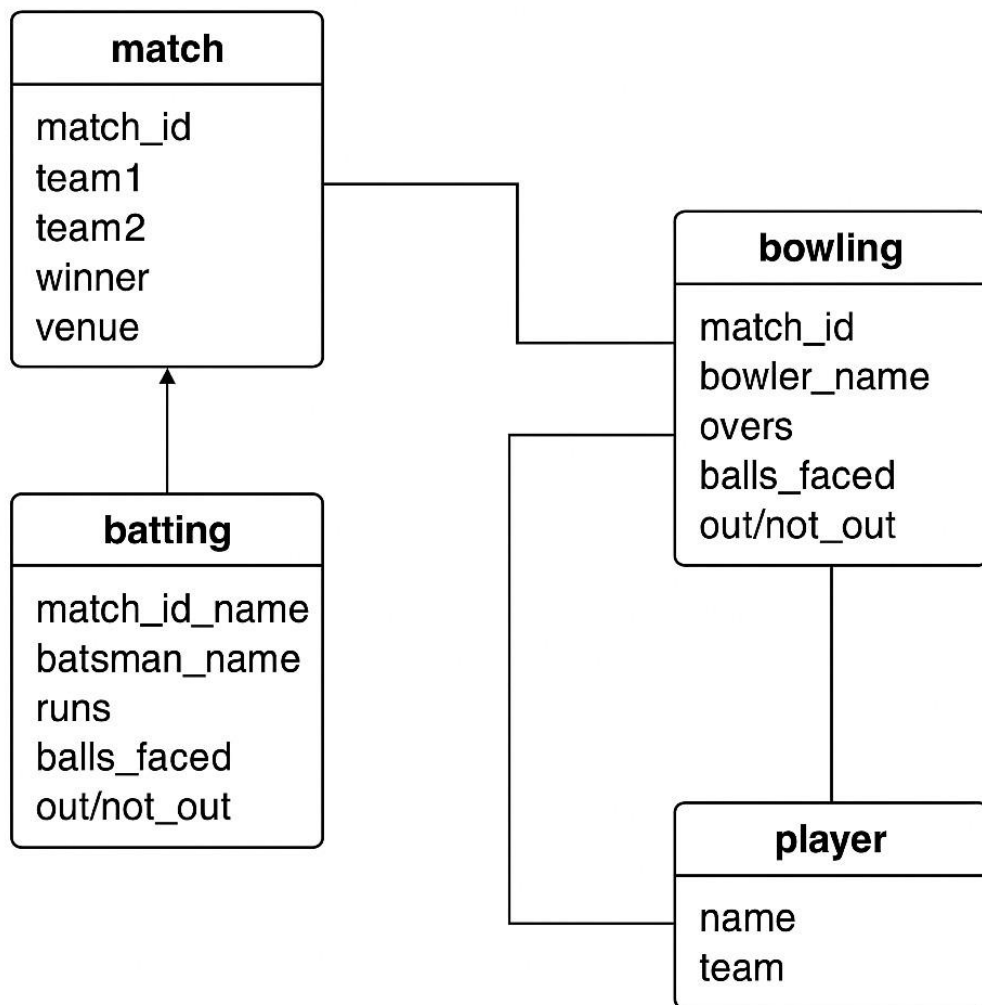
Step 7: Validation and Visualization

The selected team and rankings are visualized using graphs and tables, providing transparency in selection decisions. The team's aggregate performance metrics are compared against real-world team compositions to validate the effectiveness of the algorithm.

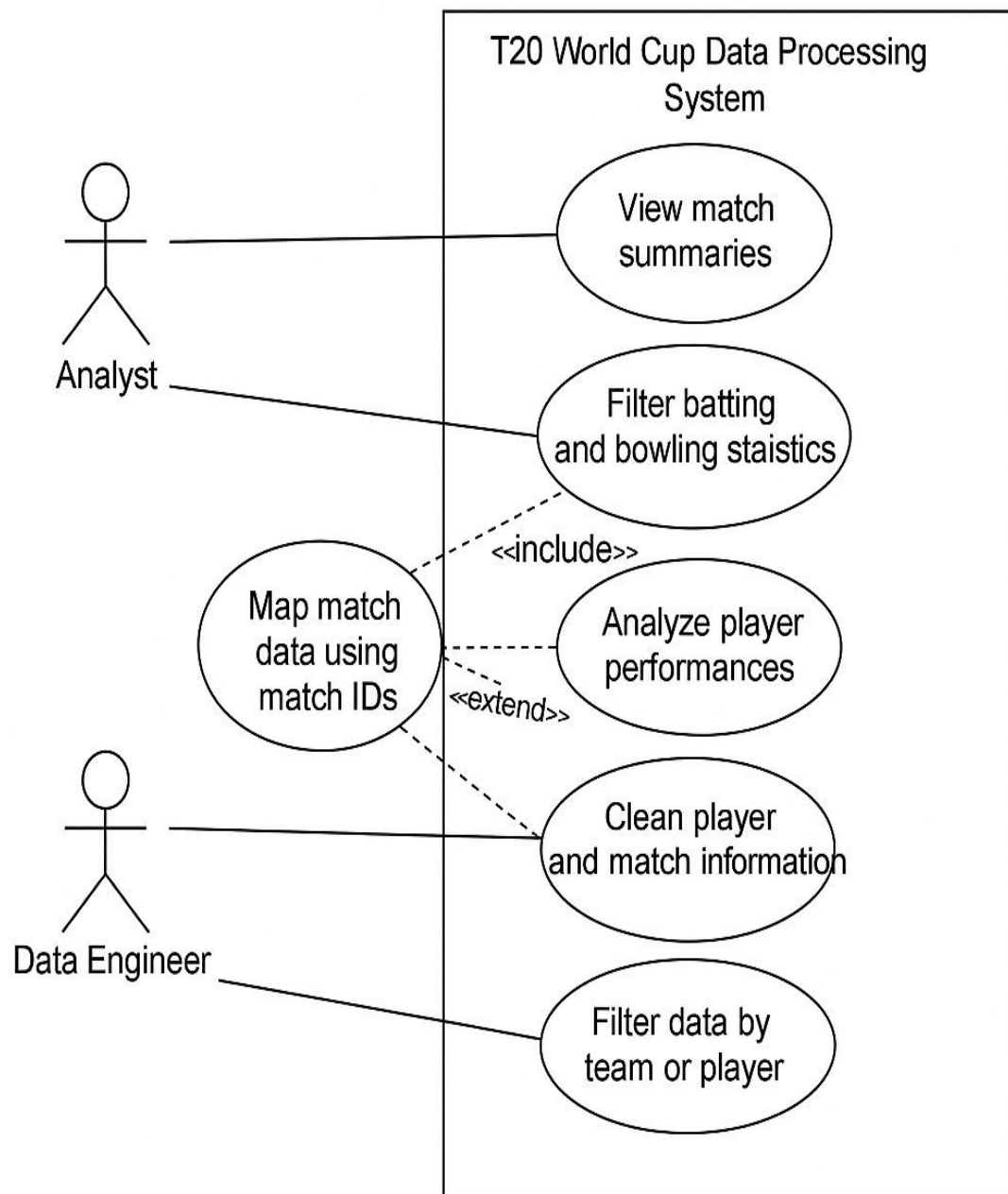
This algorithm ensures an objective, data-driven, and role-specific approach to team selection, minimizing bias and optimizing team balance for competitive success in T20 cricket.

CHAPTER 6 DESIGN FRAMEWORK

6.1 ER DIAGRAM

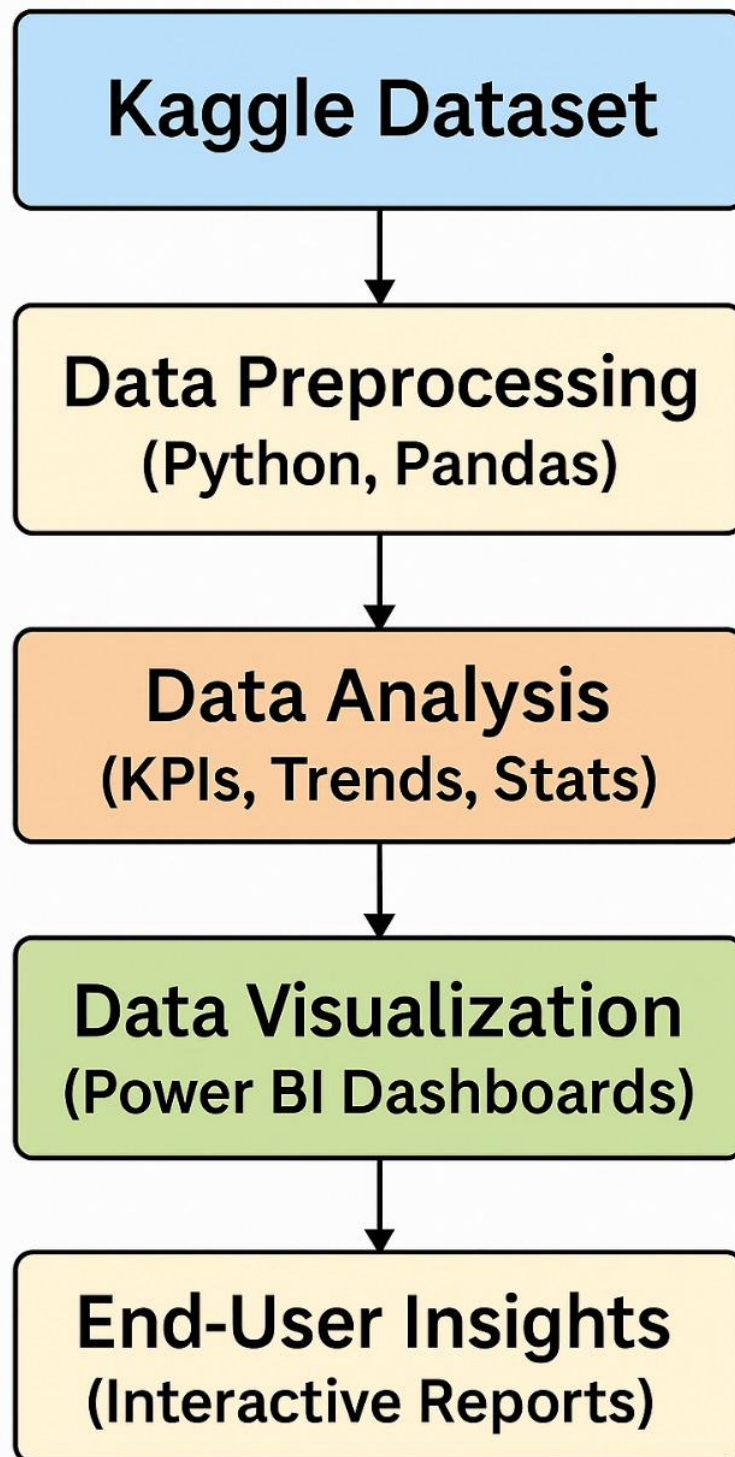


6.2 USE CASE DIAGRAM

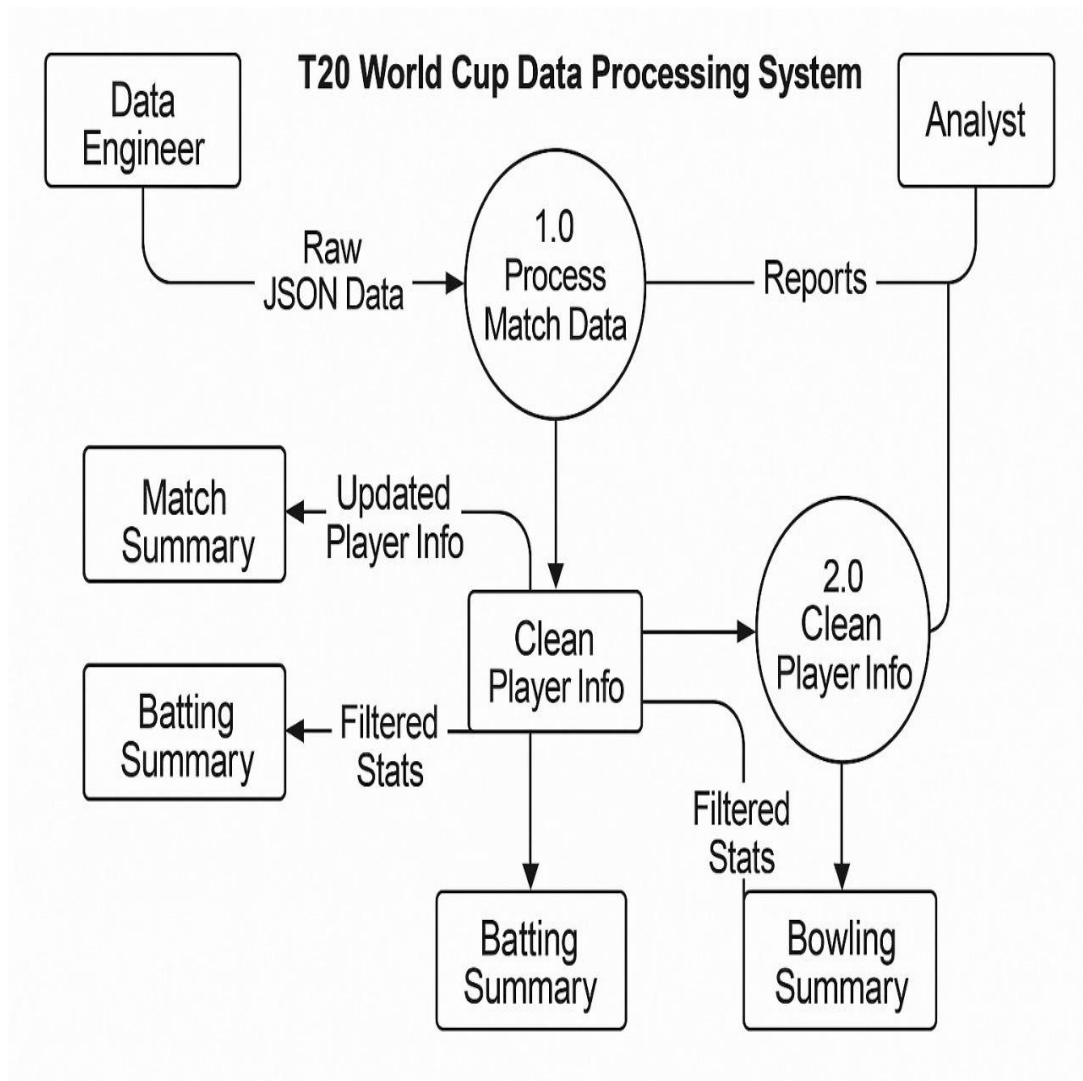


6.3 DATA FLOW DIAGRAM

0 LEVEL DATA FLOW DIAGRAM



1 LEVEL DATA FLOW DIAGRAM



CHAPTER 7 IMPLEMENTATION

7.1 SNAPSHOT

Screenshots of code and results from the notebook, including output of feature extraction, model training, and predictions, are provided below:

```
with open('/content/t20_wc_match_results.json') as f:
    data = json.load(f)

df_match = pd.DataFrame(data[0]['matchSummary'])
df_match.head()
```

	team1	team2	winner	margin	ground	matchDate	scorecard
0	Namibia	Sri Lanka	Namibia	55 runs	Geelong	Oct 16, 2022	T20I # 1823
1	Netherlands	U.A.E.	Netherlands	3 wickets	Geelong	Oct 16, 2022	T20I # 1825
2	Scotland	West Indies	Scotland	42 runs	Hobart	Oct 17, 2022	T20I # 1826
3	Ireland	Zimbabwe	Zimbabwe	31 runs	Hobart	Oct 17, 2022	T20I # 1828
4	Namibia	Netherlands	Netherlands	5 wickets	Geelong	Oct 18, 2022	T20I # 1830

```
import pandas as pd
import json
```

```
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2	Scotland	West Indies	Scotland	42 runs	Hobart	Oct 17, 2022	T20I # 1826
3	Ireland	Zimbabwe	Zimbabwe	31 runs	Hobart	Oct 17, 2022	T20I # 1828
4	Namibia	Netherlands	Netherlands	5 wickets	Geelong	Oct 18, 2022	T20I # 1830

```
[ ] df_match.shape
```

```
⇒ (45, 7)
```

```
df_match.rename({'scorecard': 'match_id'}, axis = 1, inplace = True)
df_match.head()
```

	team1	team2	winner	margin	ground	matchDate	match_id
0	Namibia	Sri Lanka	Namibia	55 runs	Geelong	Oct 16, 2022	T20I # 1823
1	Netherlands	U.A.E.	Netherlands	3 wickets	Geelong	Oct 16, 2022	T20I # 1825
2	Scotland	West Indies	Scotland	42 runs	Hobart	Oct 17, 2022	T20I # 1826
3	Ireland	Zimbabwe	Zimbabwe	31 runs	Hobart	Oct 17, 2022	T20I # 1828
4	Namibia	Netherlands	Netherlands	5 wickets	Geelong	Oct 18, 2022	T20I # 1830

```
with open('/content/t20_wc_batting_summary.json') as f:
    data = json.load(f)
    all_records = []
    for rec in data:
        all_records.extend(rec['battingSummary'])

df_batting = pd.DataFrame(all_records)
df_batting.head(11)
```

	match	teamInnings	battingPos	batsmanName	dismissal	runs	balls	4s	6s	SR
0	Namibia Vs Sri Lanka	Namibia	1	Michael van Lingen	c Pramod Madushan b Chameera	3	6	0	0	50.00
1	Namibia Vs Sri Lanka	Namibia	2	Divan la Cock	c Shanaka b Pramod Madushan	9	9	1	0	100.00
2	Namibia Vs Sri Lanka	Namibia	3	Jan Nicol Loftie-Eaton	c †Mendis b Karunaratne	20	12	1	2	166.66
3	Namibia Vs Sri Lanka	Namibia	4	Stephan Baard	c DM de Silva b Pramod Madushan	26	24	2	0	108.33
4	Namibia Vs Sri Lanka	Namibia	5	Gerhard Erasmus(c)	c Gunathilaka b PWH de Silva	20	24	0	0	83.33
5	Namibia Vs Sri Lanka	Namibia	6	Jan Frylinck	run out (Gunathilaka/†Mendis)	44	28	4	0	157.14
6	Namibia Vs Sri Lanka	Namibia	7	David Wiese	c †Mendis b Theekshana	0	1	0	0	0.00
7	Namibia Vs Sri Lanka	Namibia	8	JJ Smit		31	16	2	2	193.75
8	Namibia Vs Sri Lanka	Sri Lanka	1	Pathum Nissanka	c Smit b Shikongo	9	10	1	0	90.00
9	Namibia Vs Sri Lanka	Sri Lanka	2	Kusal Mendis†	c †Green b Wiese	6	6	0	0	100.00
10	Namibia Vs Sri Lanka	Sri Lanka	3	Dhananjaya de Silva	c Shikongo b Frylinck	12	11	1	0	109.09

```
df_batting['out/not_out'] = df_batting.dismissal.apply(lambda x: "out" if len(x)>0 else "not_out")
df_batting.head(11)
```

	match	teamInnings	battingPos	batsmanName		dismissal	runs	balls	4s	6s	SR	out/not_out
0	Namibia Vs Sri Lanka	Namibia	1	Michael van Lingen	c Pramod Madushan b Chameera		3	6	0	0	50.00	out
1	Namibia Vs Sri Lanka	Namibia	2	Divan la Cock	c Shanaka b Pramod Madushan		9	9	1	0	100.00	out
2	Namibia Vs Sri Lanka	Namibia	3	Jan Nicol Loftie-Eaton	c †Mendis b Karunaratne		20	12	1	2	166.66	out
3	Namibia Vs Sri Lanka	Namibia	4	Stephan Baard	c DM de Silva b Pramod Madushan		26	24	2	0	108.33	out
4	Namibia Vs Sri Lanka	Namibia	5	Gerhard Erasmus(c)	c Gunathilaka b PWH de Silva		20	24	0	0	83.33	out
5	Namibia Vs Sri Lanka	Namibia	6	Jan Frylinck	run out (Gunathilaka)†Mendis		44	28	4	0	157.14	out
6	Namibia Vs Sri Lanka	Namibia	7	David Wiese	c †Mendis b Theekshana		0	1	0	0	0.00	out
7	Namibia Vs Sri Lanka	Namibia	8	JJ Smit			31	16	2	2	193.75	not_out
8	Namibia Vs Sri Lanka	Sri Lanka	1	Pathum Nissanka	c Smit b Shikongo		9	10	1	0	90.00	out
9	Namibia Vs Sri Lanka	Sri Lanka	2	Kusal Mendis†	c †Green b Wiese		6	6	0	0	100.00	out
10	Namibia Vs Sri Lanka	Sri Lanka	3	Dhananjaya de Silva	c Shikongo b Frylinck		12	11	1	0	109.09	out

```
df_batting.drop(columns=["dismissal"], inplace=True)
df_batting.head(10)
```

	match	teamInnings	battingPos	batsmanName	runs	balls	4s	6s	SR	out/not_out
0	Namibia Vs Sri Lanka	Namibia	1	Michael van Lingen	3	6	0	0	50.00	out
1	Namibia Vs Sri Lanka	Namibia	2	Divan la Cock	9	9	1	0	100.00	out
2	Namibia Vs Sri Lanka	Namibia	3	Jan Nicol Loftie-Eaton	20	12	1	2	166.66	out
3	Namibia Vs Sri Lanka	Namibia	4	Stephan Baard	26	24	2	0	108.33	out
4	Namibia Vs Sri Lanka	Namibia	5	Gerhard Erasmus(c)	20	24	0	0	83.33	out
5	Namibia Vs Sri Lanka	Namibia	6	Jan Frylinck	44	28	4	0	157.14	out
6	Namibia Vs Sri Lanka	Namibia	7	David Wiese	0	1	0	0	0.00	out
7	Namibia Vs Sri Lanka	Namibia	8	JJ Smit	31	16	2	2	193.75	not_out
8	Namibia Vs Sri Lanka	Sri Lanka	1	Pathum Nissanka	9	10	1	0	90.00	out
9	Namibia Vs Sri Lanka	Sri Lanka	2	Kusal Mendis†	6	6	0	0	100.00	out

```
df_batting['match_id'] = df_batting['match'].map(match_ids_dict)
df_batting.head()
```

	match	teamInnings	battingPos	batsmanName	runs	balls	4s	6s	SR	out/not_out	match_id
0	Namibia Vs Sri Lanka	Namibia	1	Michael van Lingen	3	6	0	0	50.00	out	T20I # 1823
1	Namibia Vs Sri Lanka	Namibia	2	Divan la Cock	9	9	1	0	100.00	out	T20I # 1823
2	Namibia Vs Sri Lanka	Namibia	3	Jan Nicol Loftie-Eaton	20	12	1	2	166.66	out	T20I # 1823
3	Namibia Vs Sri Lanka	Namibia	4	Stephan Baard	26	24	2	0	108.33	out	T20I # 1823
4	Namibia Vs Sri Lanka	Namibia	5	Gerhard Erasmus(c)	20	24	0	0	83.33	out	T20I # 1823

▼ Cleanup weird characters

```
df_batting['batsmanName'] = df_batting['batsmanName'].apply(lambda x: x.replace('â€', ''))
df_batting['batsmanName'] = df_batting['batsmanName'].apply(lambda x: x.replace('\xa0', ''))
df_batting.head()
```

	match	teamInnings	battingPos	batsmanName	runs	balls	4s	6s	SR	out/not_out	match_id
0	Namibia Vs Sri Lanka	Namibia	1	Michael van Lingen	3	6	0	0	50.00	out	T20I # 1823
1	Namibia Vs Sri Lanka	Namibia	2	Divan la Cock	9	9	1	0	100.00	out	T20I # 1823
2	Namibia Vs Sri Lanka	Namibia	3	Jan Nicol Loftie-Eaton	20	12	1	2	166.66	out	T20I # 1823
3	Namibia Vs Sri Lanka	Namibia	4	Stephan Baard	26	24	2	0	108.33	out	T20I # 1823
4	Namibia Vs Sri Lanka	Namibia	5	Gerhard Erasmus(c)	20	24	0	0	83.33	out	T20I # 1823

```
with open('/content/t20_wc_bowling_summary.json') as f:
    data = json.load(f)
    all_records = []
    for rec in data:
        all_records.extend(rec['bowlingSummary'])
    all_records[:2]
```

```
[{'match': 'Namibia Vs Sri Lanka',
  'bowlingTeam': 'Sri Lanka',
  'bowlerName': 'Maheesh Theekshana',
  'overs': '4',
  'maiden': '0',
  'runs': '23',
  'wickets': '1',
  'economy': '5.75',
  '0s': '7',
  '4s': '0',
  '6s': '0',
  'wides': '2',
  'noBalls': '0'},
 {'match': 'Namibia Vs Sri Lanka',
  'bowlingTeam': 'Sri Lanka',
  'bowlerName': 'Dushmantha Chameera',
  'overs': '4',
  'maiden': '0',
  'runs': '39',
  'wickets': '1',
  'economy': '9.75',
  '0s': '6',
  '4s': '3',
  '6s': '1',
  'wides': '2',
  'noBalls': '0'}]
```

```
df_bowling = pd.DataFrame(all_records)
print(df_bowling.shape)
df_bowling.head()
```

(500, 13)

	match	bowlingTeam	bowlerName	overs	maiden	runs	wickets	economy	0s	4s	6s	wides	noBalls
0	Namibia Vs Sri Lanka	Sri Lanka	Maheesh Theekshana	4	0	23	1	5.75	7	0	0	2	0
1	Namibia Vs Sri Lanka	Sri Lanka	Dushmantha Chameera	4	0	39	1	9.75	6	3	1	2	0
2	Namibia Vs Sri Lanka	Sri Lanka	Pramod Madushan	4	0	37	2	9.25	6	3	1	0	0
3	Namibia Vs Sri Lanka	Sri Lanka	Chamika Karunaratne	4	0	36	1	9.00	7	3	1	1	0
4	Namibia Vs Sri Lanka	Sri Lanka	Wanindu Hasaranga de Silva	4	0	27	1	6.75	8	1	1	0	0

```
[ ] df_bowling['match_id'] = df_bowling['match'].map(match_ids_dict)
df_bowling.head()
```

(500, 14)

	match	bowlingTeam	bowlerName	overs	maiden	runs	wickets	economy	0s	4s	6s	wides	noBalls	match_id
0	Namibia Vs Sri Lanka	Sri Lanka	Maheesh Theekshana	4	0	23	1	5.75	7	0	0	2	0	T20I # 1823
1	Namibia Vs Sri Lanka	Sri Lanka	Dushmantha Chameera	4	0	39	1	9.75	6	3	1	2	0	T20I # 1823
2	Namibia Vs Sri Lanka	Sri Lanka	Pramod Madushan	4	0	37	2	9.25	6	3	1	0	0	T20I # 1823
3	Namibia Vs Sri Lanka	Sri Lanka	Chamika Karunaratne	4	0	36	1	9.00	7	3	1	1	0	T20I # 1823
4	Namibia Vs Sri Lanka	Sri Lanka	Wanindu Hasaranga de Silva	4	0	27	1	6.75	8	1	1	0	0	T20I # 1823

```
] with open('/content/t20_wc_player_info.json') as f:
    data = json.load(f)
```

```
df_players = pd.DataFrame(data)
```

```
print(df_players.shape)
df_players.head(10)
```

(219, 6)

	name	team	battingStyle	bowlingStyle	playingRole	description
0	Michael van Lingen	Namibia	Left hand Bat	Left arm Medium	Bowling Allrounder	
1	Divan la Cock	Namibia	Right hand Bat	Legbreak	Opening Batter	
2	Jan Nicol Loftie-Eaton	Namibia	Left hand Bat	Right arm Medium, Legbreak	Batter	
3	Stephan Baard	Namibia	Right hand Bat	Right arm Medium fast	Batter	
4	Gerhard Erasmus(c)	Namibia	Right hand Bat	Right arm Offbreak	Allrounder	
5	Jan Frylinck	Namibia	Left hand Bat	Left arm Fast medium	Allrounder	
6	David Wiese	Namibia	Right hand Bat	Right arm Medium fast	Allrounder	David Wiese joined a marked outflow of South A...
7	JJ Smit	Namibia	Right hand Bat	Left arm Medium fast	Bowling Allrounder	
8	Pathum Nissanka	Sri Lanka	Right hand Bat		Top order Batter	
9	Kusal Mendis†	Sri Lanka	Right hand Bat	Legbreak	Wicketkeeper Batter	Blessed with a compact technique, an aggressiv...

Cleanup weird characters

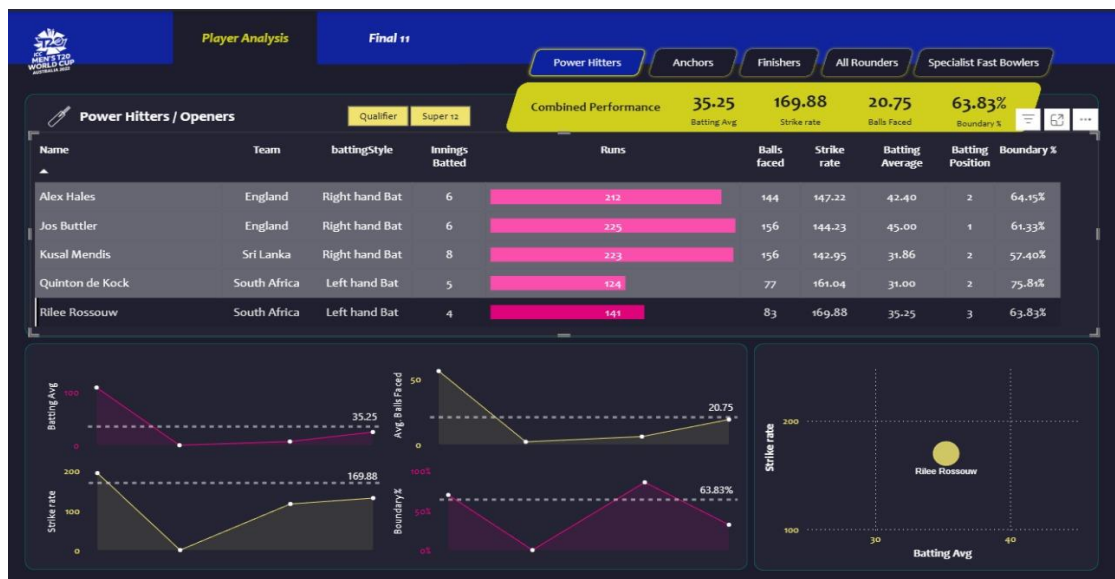
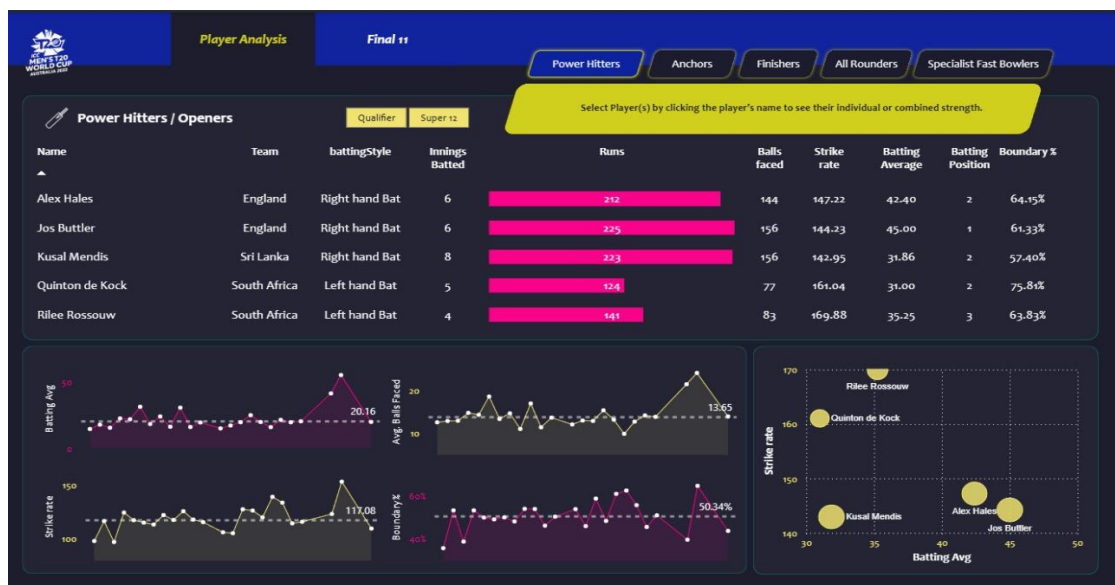
```
df_players['name'] = df_players['name'].apply(lambda x: x.replace('â€', ''))
df_players['name'] = df_players['name'].apply(lambda x: x.replace('†', ''))
df_players['name'] = df_players['name'].apply(lambda x: x.replace('\xa0', ''))
df_players.head(10)
```

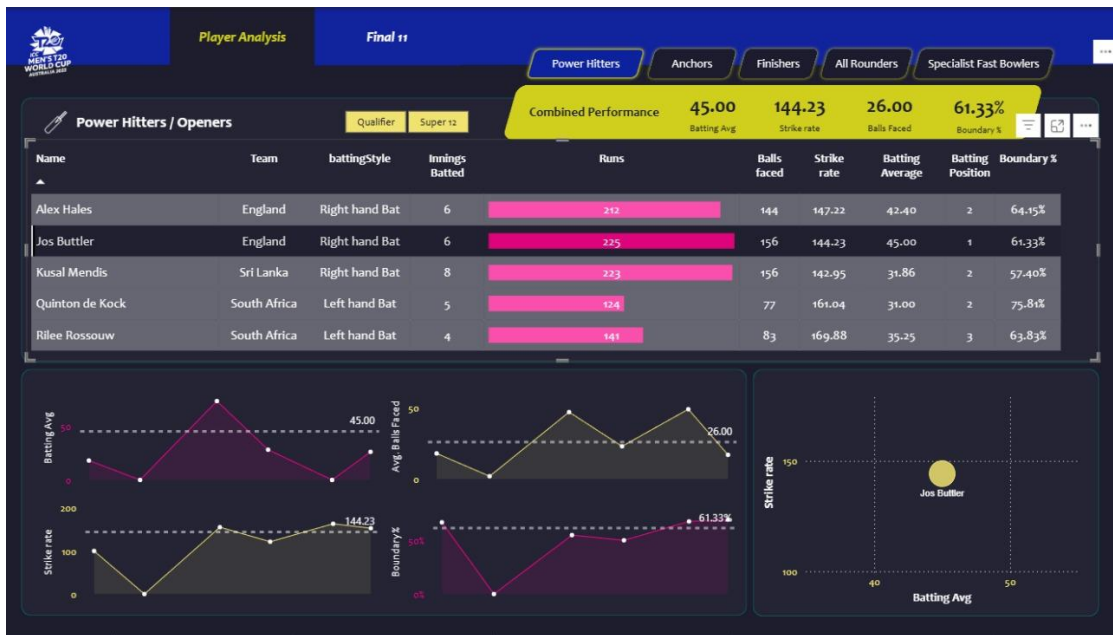
(219, 6)

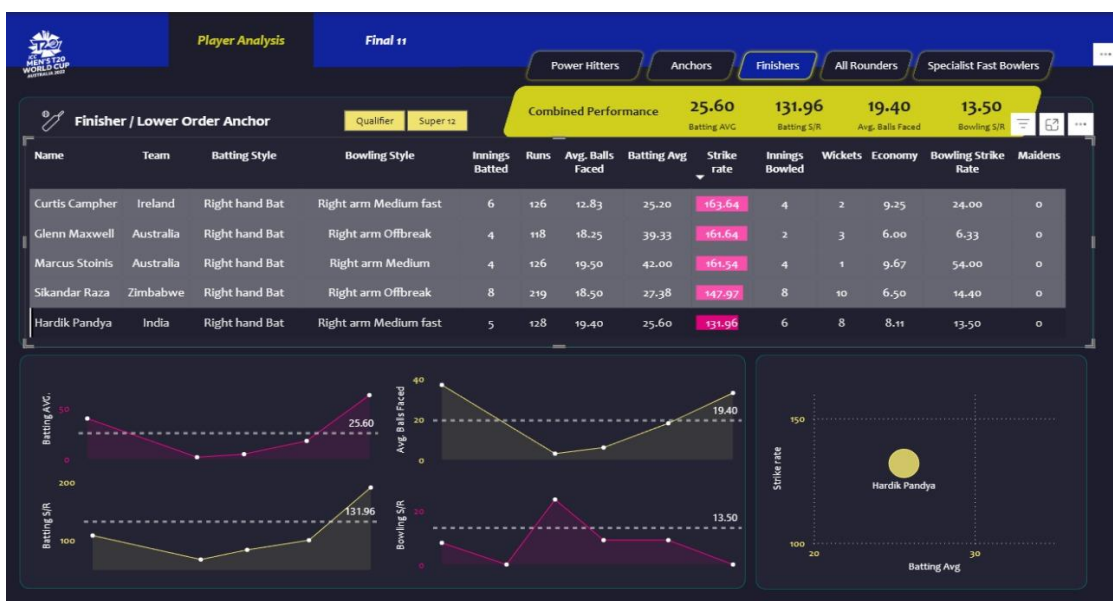
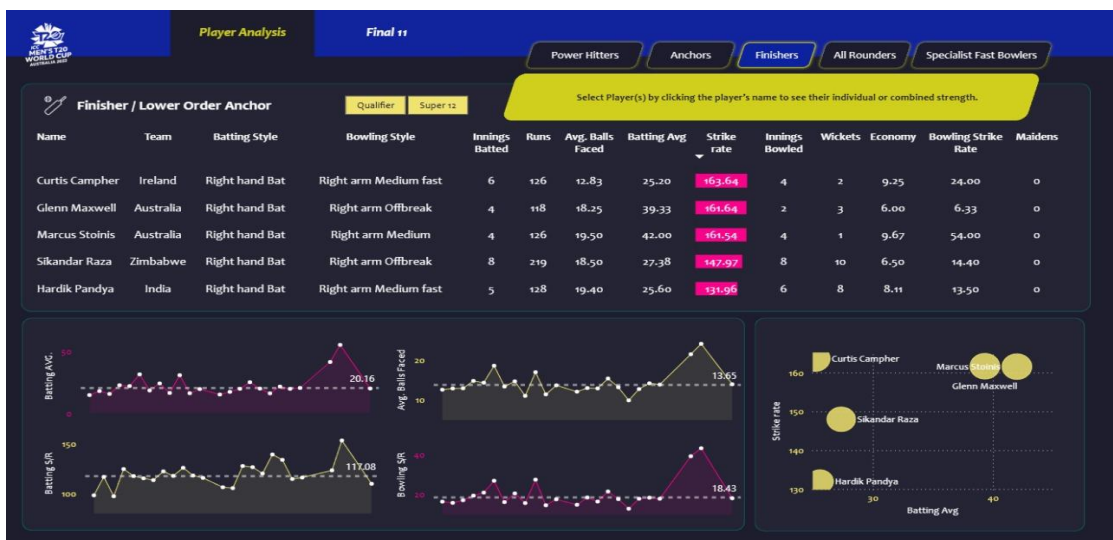
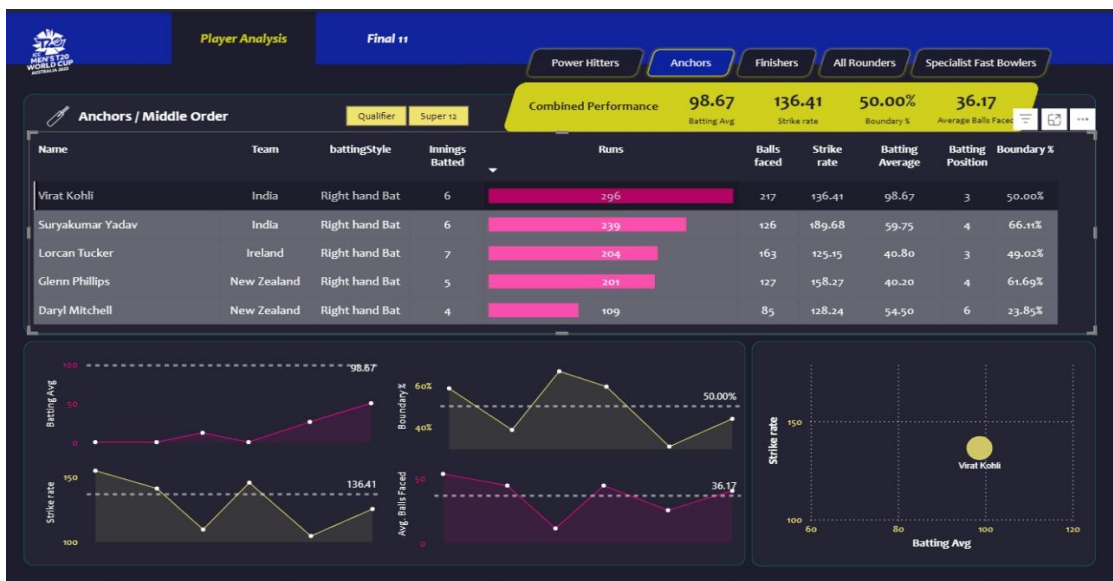
	name	team	battingStyle	bowlingStyle	playingRole	description
0	Michael van Lingen	Namibia	Left hand Bat	Left arm Medium	Bowling Allrounder	
1	Divan la Cock	Namibia	Right hand Bat	Legbreak	Opening Batter	
2	Jan Nicol Loftie-Eaton	Namibia	Left hand Bat	Right arm Medium, Legbreak	Batter	
3	Stephan Baard	Namibia	Right hand Bat	Right arm Medium fast	Batter	
4	Gerhard Erasmus(c)	Namibia	Right hand Bat	Right arm Offbreak	Allrounder	
5	Jan Frylinck	Namibia	Left hand Bat	Left arm Fast medium	Allrounder	
6	David Wiese	Namibia	Right hand Bat	Right arm Medium fast	Allrounder	David Wiese joined a marked outflow of South A...
7	JJ Smit	Namibia	Right hand Bat	Left arm Medium fast	Bowling Allrounder	
8	Pathum Nissanka	Sri Lanka	Right hand Bat		Top order Batter	
9	Kusal Mendis	Sri Lanka	Right hand Bat	Legbreak	Wicketkeeper Batter	Blessed with a compact technique, an aggressiv...

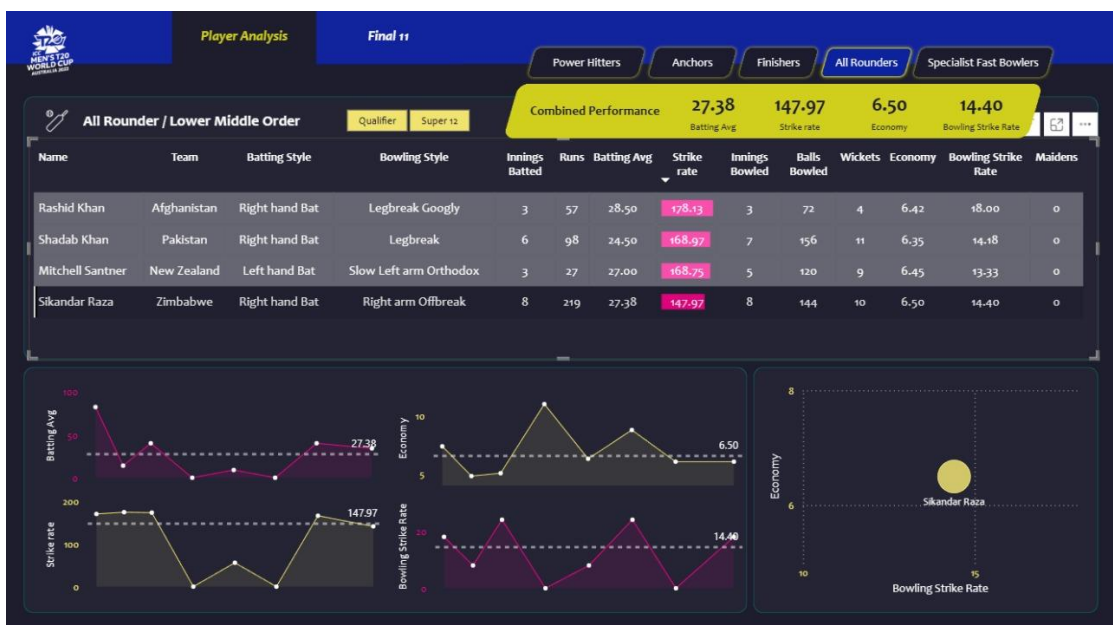
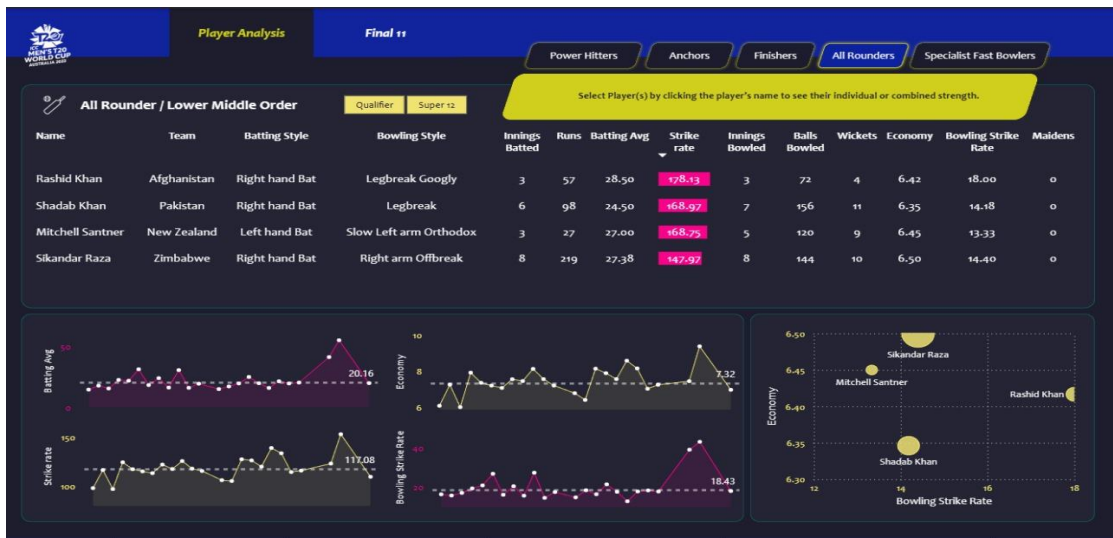
```
df_players[df_players['team'] == 'India']
```

	name	team	battingStyle	bowlingStyle	playingRole	description
127	KL Rahul	India	Right hand Bat		Opening Batter	A tall, elegant right-hand batsman who can kee...
128	Rohit Sharma(c)	India	Right hand Bat	Right arm Offbreak	Top order Batter	Languid and easy on the eye, Rohit Sharma owne...
129	Virat Kohli	India	Right hand Bat	Right arm Medium	Top order Batter	India has given to the world many a great cric...
130	Suryakumar Yadav	India	Right hand Bat	Right arm Medium, Right arm Offbreak	Batter	Hard-hitting 360-degree batter Suryakumar Yada...
131	Axar Patel	India	Left hand Bat	Slow Left arm Orthodox	Bowling Allrounder	Left-arm spinner Axar Patel has been increas...
132	Hardik Pandya	India	Right hand Bat	Right arm Medium fast	Allrounder	Hardik Pandya swears by living life king size ...
133	Dinesh Karthik	India	Right hand Bat	Right arm Offbreak	Wicketkeeper Batter	Not many would forget the sight of Dinesh Kart...
134	Ravichandran Ashwin	India	Right hand Bat	Right arm Offbreak	Bowling Allrounder	R Ashwin took the tricks and skills he learned...
135	Bhuvneshwar Kumar	India	Right hand Bat	Right arm Medium	Bowler	At the time of his India debut in 2012, Bhuvne...
136	Arshdeep Singh	India	Left hand Bat	Left arm Medium fast	Bowler	
137	Mohammed Shami	India	Right hand Bat	Right arm Fast	Bowler	Mohammed Shami was India's leading fast bowler...
192	Deepak Hooda	India	Right hand Bat	Right arm Offbreak	Allrounder	An allrounder who can bat in any position, Dee...
211	Rishabh Pant	India	Left hand Bat		Wicketkeeper Batter	A match-turning, swashbuckling batter-keeper i...









CHAPTER 8 TESTING

8.1 TESTING

Player's Name	Image	Team	Batting Style	Playing Role	Bowling Style	Batting AVG.	Batting S/R	Economy	Bowling S/R	Bowling AVG.	Custom Batting Order
Jos Buttler		England	Right hand Bat	Wicketkeeper Batter		45.00	144.33				1
Rilee Rossouw		South Africa	Left hand Bat	Top order Batter	Right arm Offbreak	35.25	169.88				2
Virat Kohli		India	Right hand Bat	Top order Batter	Right arm Medium	98.67	136.41				3
Suryakumar Yadav		India	Right hand Bat	Batter	Right arm Medium, Right arm Offbreak	59.75	189.68				4
Glenn Phillips		New Zealand	Right hand Bat	Wicketkeeper Batter	Right arm Offbreak	40.20	158.27				5

Team Performance	40.49	154.96	20.38	15.46	13.93	6.66	38.85%
	Batting Avg	Strike rate	Average Balls Faced	Bowling Avg	Bowling S/R	Economy	Dot ball %

Testing is a crucial phase in any data-driven project, ensuring the accuracy, consistency, and reliability of the outcomes. In this project, testing was performed at multiple stages to validate data processing, transformation logic, and selection accuracy of the top 15 players.

The first step of testing involved verifying the integrity of the datasets collected from ESPNcricinfo. JSON files for match summaries, batting, bowling, and player info were checked for completeness and consistency. Data cleaning functions were tested to ensure the removal of unwanted characters, proper mapping of match IDs, and elimination of null or duplicate entries.

Functional testing was applied to key logic areas like classification of players (batter, bowler, opener, all-rounder), calculation of performance metrics (e.g., strike rate, batting average, economy rate), and mapping match data to players. Pandas assertions and manual sampling were used to verify correct calculations.

Additionally, unit testing was done on functions responsible for ranking and sorting players by category. Visual outputs like graphs and team composition tables were cross-checked for accuracy and alignment with expected results.

CHAPTER 9 CONCLUSION & FUTURE SCOPE

9.1 CONCLUSION & FUTURE SCOPE

This project successfully demonstrates the application of data analytics techniques to cricket, specifically the ICC T20 World Cup 2022. By gathering real-time match data from ESPNcricinfo and processing player-wise summaries for batting, bowling, and team compositions, we were able to build a data-driven model to evaluate player performance. Our aim was to construct an optimal team of 15 players—including openers, batters, bowlers, and all-rounders—based on role-specific performance metrics and criteria.

The project began by preprocessing multiple datasets: match summaries, player information, batting and bowling statistics. These datasets were cleaned, normalized, and mapped using consistent match identifiers to ensure accurate linking of performances to players and matches. Players were then categorized into their respective roles based on their match behavior—batting positions and bowling contributions. Metrics such as batting average, strike rate, bowling economy, and all-round performance indices were calculated and used to rank players objectively within their respective categories.

The methodology enabled the fair selection of top-performing players, ensuring a balanced and competitive squad. The use of statistical methods provided transparency and removed subjective bias from team formation, highlighting the power of analytics in modern sports decision-making. Furthermore, data visualizations supported the analysis, offering insights into player trends, match outcomes, and team strengths.

In terms of significance, this project not only aids selectors and coaches in decision-making but also benefits sports analysts and fans who wish to understand player impact through data. It showcases how sports data can be converted into actionable intelligence for better team strategy and management.

Future Scope

The scope for future enhancement is vast. Firstly, the algorithm can be extended by incorporating machine learning models to predict future player performance based on historical data, form, and opponent statistics. Advanced analytics such as player momentum, fitness data, and opposition analysis can be integrated to make the selection process even more comprehensive.

Real-time streaming data from live matches could be processed for dynamic team adjustments. Additionally, similar models can be adapted for ODI and Test formats, or even domestic leagues like IPL, BBL, etc. Further, the project can be converted into a web application or dashboard to allow users to generate teams interactively using custom filters and preferences.

In conclusion, this project is a powerful demonstration of how data science can transform cricket analysis, enabling smarter, data-informed team strategies. The integration of more real-time and granular data sources will only make such systems more accurate and impactful in the future.

REFERENCES

1. ESPNcricinfo – T20 World Cup 2022 Data

Description:

Official statistics, scorecards, player details, and match summaries of all ICC T20 World Cup 2022 matches. Used as the primary data source for this project.

Website Link:

<https://www.espnricinfo.com>

2. A Data-Driven Approach to Performance Analysis in T20 Cricket – IEEE Conference

Description:

This paper presents methods for analyzing T20 player performance using structured data and statistical metrics. Helpful for understanding role-specific evaluation.

Website Link:

<https://ieeexplore.ieee.org/document/9316026>

3. Player Performance Prediction and Ranking in T20 Cricket – IJACSA Journal

Description:

Focuses on ranking cricket players using machine learning and statistical measures. Provided insight into ranking algorithms used in this project.

Website Link:

https://thesai.org/Downloads/Volume11No4/Paper_44-Player_Performance_Prediction_and_Ranking.pdf

4. Sports Analytics in Cricket: A Review on Performance Evaluation Techniques

Description:

A literature review paper that surveys various performance evaluation techniques in cricket. Helped build the theoretical foundation of the project.

Website Link:

https://www.researchgate.net/publication/352734964_Sports_Analytics_in_Cricket

5. Use of Machine Learning for Predicting Outcomes in T20 Cricket – IJCA

Description:

Explores prediction and player performance using ML algorithms. Inspired the idea of using weighted metrics for all-rounder evaluations.

Website Link:

<https://www.ijcaonline.org/archives/volume177/number8/30536-2019919494>

6. Cricket Analytics: The Numbers Game Behind the Game – Book by J. Miller

Description:

A comprehensive book on sports analytics applied to cricket. Useful for understanding data handling and cricket-specific performance indicators.

Website Link:

<https://link.springer.com/book/10.1007/978-3-319-94336-8>